

Instruction to student:.

1. Attempt any five Questions from Question No. 1 to Question No.7.
2. Each and every question carries equal mark
3. Illustrate your answers with neat sketches, diagrams etc. wherever necessary.

Q.1 What are the different types of error in numerical methods and Explain how error propagates in numerical method through division and multiplication?

Q.2) Solve the following system of equations up to correct two decimal places by (i) Gauss-Seidel Method (ii) Gauss-Jacobi method : $28x + 4y - z = 32$; $x + 3y + 10z = 24$; $2x + 17y + 4z = 35$

Q.3) (a) What is function prototyping explain in brief?

(b) Comment on declaration of variable in C++ and explain Dynamic initialization.

Q.4) Find a root of the equation $x^3 + 3x - 5 = 0$ upto correct five decimal places using newton-raphson method with initial value $x_0 = 1$.

Q.5) Use Runge- Kutta Method of fourth order to find y_1 at $x = 0.1$ and y_2 at $x = 0.2$, given that

$$\frac{dy}{dx} = 1 + xy ; y(0) = 2$$

Q.6) Write short note on (Any two)

- a) Classes and objects b) Object oriented Programming c) Operators in C++

Q.7 (a) Use lagranges formula to find value of y at $x = 2$:

x	0	1	3	4
y	-12	0	6	12

(b) Explain finite difference operator i.e forward and backward difference and their properties in detail.

Q.8 (a) Evaluate $\Delta^2 f(x)$ where $f(x) = (5x + 12) / (x^2 + 5x + 6)$ taking interval difference as $h = 1$.

(b) Evaluate integral of $dx / (1 + x^2)$ within limits (0,10) by using simpsons one third rule. Take h as 1

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